Exercises on Magmas, Monoids and Groups (Algebra I)

Sept 13, 2026

1. Which of the following laws of composition are associative or commutative?

   (a) On the set $\mathbb{Z}$ of integers, $a * b = (a+3)(b+2)$ $\forall a, b \in \mathbb{Z}$.

   (b) On the set $\mathbb{R}$, $a * b = a + ab$ $\forall a, b \in \mathbb{R}$.

   (c) On the power set $\mathcal{P}(X)$ of a set X,

   $$A * B = \mathcal{C}_X(A) \cup \mathcal{C}_X(B) \quad \forall A, B \in \mathcal{P}(X).$$

   (d) On any set X,

   $$a * b = a \quad \forall a, b \in X.$$

2. Prove or disprove the statement: "Every commutative law of composition on a 2-element set is associative."

3. Determine the number of laws of composition and commutative laws of composition on a set with n elements.

4. Let $(S, *)$ be an associative magma and e be any element not in S and $S' = S \cup \{e\}$. Define a law of composition $+$ on S' as follows. For any $a, b \in S'$, define

   $$a + b = \begin{cases} a * b & \text{if both } a \text{ and } b \in S, \\ a & \text{if } b = e, \\ b & \text{if } a = e. \end{cases}$$

   Then prove that $(S', +)$ is a monoid with neutral element e.

5. Which of the following magmas is a group?

   (a) $(\mathbb{Z}, *)$, where $a * b = a + b + ab$ $\forall a, b \in \mathbb{Z}$.

   (b) $(\mathbb{R}_{>0}, *)$, where $a * b = \dfrac{a}{b}$ $\forall a, b \in \mathbb{R}_{>0}$ and $\mathbb{R}_{>0}$ denote the set of positive real numbers.

6. Let n be a positive integer and G be the set of all nth roots of unity; that is $G = \{z : z \text{ is a complex number and } z^n = 1\}$. Prove that G is a group under the usual multiplication of complex numbers.

7. Let $(G, *)$ be a group and X be any nonempty set. Let G^X be the set of all mappings of X into G. For any $f, g \in G^X$, define $f * g : X \to G$ by

$$(f * g)(x) = f(x) * g(x) \quad \forall x \in X.$$

Prove that $(G^X, *)$ is a group.